Modern Physics

Introduction to Statistical Mechanics

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Foreword

Why should we learn Statistical Mechanics?

- From Physics to Biology, from Chemistry to Materials, it is seen that new laws can emerge when the number of particles becomes large, which are different from the laws followed by the individual particles. Statistical Mechanics forms the basis of our understanding of how this can happen.
- Statistical Mechanics further clarifies the meaning of entropy in large systems. This reveals interesting connections with information theory.
- A large variety of optimization problems are most often analyzed by casting them into Statistical Mechanics problems.
- The Nobel Prize in Physics in 2024 was given to John Hopfield and Geoffrey Hinton for laying foundations of Artificial Intelligence, more specifically, neural networks. The connection between neural networks and Physics is via Statistical Mechanics.

References:

- Chapter 1 of Statistical Mechanics, R. K. Pathria.
- Chapter 5 of Introduction to Quantum Mechanics, David J. Griffiths.

[*Everything that will be discussed in this course can be found in the above books. But, it may be easier to read these notes, as they are more focussed on specific things. Further, reading the notes may become even more easy if classes are attended.*]

1 Background and the main goal of statistical mechanics

1.1 Microscopic and macroscopic views

1.1.1 Microscopic view: Newton's laws

Consider a single (classical) particle in a box that is completely isolated from the surroundings. The particle moves around either freely or under some external force (say gravity), and interacts with the walls of the box in some way (say, via elastic collisions). According to classical mechanics, the 'state' of such a system is completely defined microscopically if we know the position and the momentum of the particle. Thus, the 'state' at any time t is described in terms of 6 variables,

$$\{x(t), y(t), z(t), p_x(t), p_y(t), p_z(t)\}, \tag{1.1}$$

where $x(t), y(t), z(t)$ give the coordinates of the particle in x, y and z directions respectively, and $p_x(t), p_y(t), p_z(t)$ are the corresponding components of momentum. Given the position and momentum of the particle at any chosen time, we can predict the state of the system at any later time using Newton's laws of motion. The number of variables required to describe the state of the system microscopically is called the 'degrees of freedom'. So, here the number of degrees of freedom is 6. Since we need to determine 6 variables, we need 6 equations to solve for them. So, the degrees of freedom essentially gives the number of equations we need to solve to microscopically describe the state of the system.

If we have two such particles, we need to know the positions and momenta of both particles, so, there needs to be 12 variables, viz.,

$$\{x_1(t), y_1(t), z_1(t), p_{x_1}(t), p_{y_1}(t), p_{z_1}(t), x_2(t), y_2(t), z_2(t), p_{x_2}(t), p_{y_2}(t), p_{z_2}(t)\}, \tag{1.2}$$

where the label 1 is for first particle and label 2 is for the second particle. Given the position and the momenta for both particles at any given time, and if we know how the particles interact with each other (say, elastic collisons, or may be, even simpler, does not interact at all), the external forces on them (such as gravity) and how they interact with the walls (say, elastic collisons), we can predict the state of the system at any later time using Newton's laws. In this case the degrees of freedom is 12. So, we need to solve 12 equations microscopically describe the state of the system. This is already getting tougher, but may be still manageable.

Generalizing the above to N particles, to microscopically describe the state of a system, we need to know position and momenta of all the particles, so,

$$\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}, \quad j = 1 \text{ to } N. \tag{1.3}$$

So, the number of degrees of freedom is $6N$. A given set of values for all the degrees of freedom is called a 'microstate' (short for microscopic state) of the system.

However, most commonly, we have systems made up of a huge number of particles. To get an estimate, they are usually made of several moles of particles, and one mole is Avogadro's number ($N_A \sim 10^{24}$) of particles! Such systems are called macroscopic systems. It is completely hopeless to try to solve these many equations! In other words, although Newton's laws give us the way to in principle predict the state of the system, they are completely impractical to use for macroscopic systems!

In above, we have assumed the particles to be classical. The situation remains essentially the same (or sometimes becomes worse) if particles are quantum mechanical. Instead of Newton's laws, their degrees of freedom are to be obtained via solving the Schroedinger's equation or the Heisenberg's equations of motion. Nevertheless, the number of degrees of freedom still remains proportional to the number of particles. Hence, for macroscopic systems, any microscopic description given via specifying all degrees of freedom becomes impractical.

So, what do we do? Fortunately, the way out is provided by thermodynamics.

Key takeaways:

For a system of N particles

- 'Microstate' is given by

$$\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}, \quad j = 1 \text{ to } N.$$

- Given a microstate at any particular time, microstate at other times can be obtained via microscopic laws of motion, viz., Newton's laws [or may be Heisenberg's equaiton of motion, if particles are quantum mechanical].
- Number of variables to describe microstate = 'Degrees of freedom'. So, degrees of freedom is $6N$ [importantly, proportional to N].
- Macroscopic system $\Rightarrow$ N very large, say of the order of Avogadro's number $N_A \sim 10^{24}$.
- So, for macroscopic systems, describing the system in terms of microstates becomes utterly impractical, although, in principle, it is possible.

1.1.2 Macroscopic view: Laws of thermodynamics

Empirical observations: Macroscopic variables and macrostate —
Thermodynamics is based on the following empirical (i.e, experimental) observations.

- *Observation 1.1:* Macroscopic systems seem to be well-described by few 'macroscopic variables', say $\{\theta_j\}$, $j = 1, 2, 3 \ldots, r$. The value of n is a small tractable one. It can be 2 or 3, or may be even 10, but definitely not $\sim 10^{24}$. For example, to describe a gas, we usually need pressure P, volume V, temperature T, so $\theta_1 = P$, $\theta_2 = V$, $\theta_3 = T$, thus $r = 3$. A specified set of macroscopic variables is said to describe the 'macrostate' (short for macroscopic state) of the system.

- *Observation 1.2:* The macroscopic variables tend to remain const over quite long durations of time. Whenever all macroscopic variables are constant, the system is said to be in 'equilibrium state'.

- *Observation 1.3:* Starting from equilibrium, if a macroscopic variable (say P) is changed experimentally, one or more of the other macroscopic variables (say V, T) change quickly to different values, where again, they remain constant. We say, the system 'equilibrates' to a 'new' equilibrium state. Thus, the macroscopic variables are not independent of one another in equilibrium state. Experimentally, their functional dependence on one another can be found, formally written as $f(\{\theta_j\}) = 0$. For example, in a gas, we usually have $f(P, V, T) = 0$. Such an equation is called 'equation of state'. The most famous equation of state is, of course, the equation of state for an ideal gas: $PV - nRT = 0$, where n is number of moles of gas, R is the ideal-gas constant, and T is temperature in Kelvin.

Based on these observations, thermodynamics goes on to formulate the laws for the macroscopic variables. These laws, serve two purposes: (a) they make the meaning of heat and temperature precise, (b) they describe how the macrostate of the system evolves, much like how Newton's laws describe the evolution of microstate of the system. We will not dwell on point (a) above here, since this is not a course on thermodynamics itself. Rather, we will assume that we have at least an intuitive idea of heat and temperature. However, we will delve a bit deeper into point (b). For simplicity and concreteness, in the following, we will only consider so called, 'hydrostatic systems', i.e, systems with macroscopic variables P, V, T, keeping in mind that, all discussions can be generalized to any macroscopic system, described by any set of finite number of macroscopic variables.

First and second laws of thermodynamics —
Thermodynamics most often talks about a system coupled to its surroundings, with the surrounding being at some temperature. Imagine a situation, where the macroscopic variables of a hydrostatic system are changed from an initial to a final value, say $\{P_i, V_i, T_i\}$ to $\{P_f, V_f, T_f\}$. Initial and final states are in equilibrium, so satisfy the equation of state, i.e, $f(P_i, V_i, T_i) = 0$, $f(P_f, V_f, T_f) = 0$. The change of macrostate is governed by the most famous laws of thermodynamics, the first and the second laws [there are two more laws: the zeroth and the third laws, which hardly anyone talks about outside a Physics class]. The first law is,

$$\text{First law: } dE = đQ + đW. \tag{1.4}$$

Here $đQ$ is the differential amount of heat absorbed by the system from the surroundings during a small step of the process, $đW$ is the differential amount of work done on the system in the same step, and dE is the differential amount of change in internal energy of the system due to the heat absorbed and the work done. The notation $đQ$ implies $\int_i^f đQ \neq Q_f - Q_i$, i.e, the total heat absorbed cannot be written as 'heat of the system in final state' minus 'heat of the system in the initial state'. Similarly, the $đW$ implies

$\int_i^f đW \neq W_f - W_i$, with similar meaning of all notations. In contrast, $\int_i^f dE = E_f - E_i$, i.e, the total change in internal energy of the system is given by the internal energy of the system in the final state minus the internal energy of the system in the initial state. This guarantees that the internal energy can be expressed as a function of the macroscopic variables of the state, i.e., $E_i = E_i(P_i, V_i)$, $E_f = E_f(P_f, V_f)$, unlike heat and work. We have omitted the temperature while writing internal energy as fuction of macroscopic variables, because the equation of state can be used to express temperature as a function of volume and pressure, i.e, there are two independent variables. We could have equivalently chosen any other set of two variables, like pressure and temperature, instead.

While taking about thermodynamics, it is important to fix the sign convention for heat and work. Our convention is:

$đQ > 0 \rightarrow$ Heat absorbed by the system, $đW > 0 \rightarrow$ Work done on the system,
$đQ < 0 \rightarrow$ Heat given out by the system, $đW < 0 \rightarrow$ Work done by the system.

Having discussed the first law, we now go to the second law.

$$\text{Second law: } S_f - S_i + \int_i^f \frac{-đQ}{T} \geq 0. \tag{1.5}$$

Here, S_f is the entropy of the final state, S_i is the entropy of the final state and T is the temperature of the surroundings, which may, in principle, change during the process. In general, we will always use the letter S to denote entropy. The $-đQ$ in second term in Eq. (1.5), by our convention, denotes the differential amount of heat given out by the system during a small step of the process. The second term in Eq. (1.5) is sometimes called 'change in entropy of the bath' or 'change in entropy of the surroundings'. In this way of thinking, system+surroundings consists the 'universe', and Eq. (1.5) is thus change in entropy of the 'universe', since it is such of 'change in entropy of system' and 'change in entropy of surroundings'. So, (1.5) is the mathematical statement of the popular saying 'entropy of the universe cannot decrease'.

Reversible and irreversible processes —
Both the first law and the second law are valid for any process where the macroscopic variables change. The process can be either 'reversible' or 'irreversible'. A reversible process is one where throughout the process the system remains close to an instantaneous equilibrium state. So, starting with $\{P_i, V_i, T_i\}$ we change any two of the macroscopic variables, say, volume and temperature, by a tiny amount, and wait a little time for the remaining variable, say pressure, to change such that the equation of state is again satisfied with the new values. Then we again change make a tiny change. We keep doing this, until we reach $\{P_f, V_f, T_f\}$. This can be more formally written as follows. Discretize the interval between $\{P_i, V_i, T_i\}$ and $\{P_f, V_f, T_f\}$ into ℓ steps, such that $dP = (P_f - P_i)/\ell$, $dV = (V_f - V_i)/\ell$, $dT = (T_f - T_i)/\ell$, ℓ being large. The process occurs through these discretized steps, the pressure, volume, temperature at rth step being $P_i + rdP$, $V_i + rdV$, $T_i + rdT$. The crucial point is that the equation of state is satisfied in each of the steps,

i.e, $f(P_i + rdP, V_i + rdV, T_i + rdT) = 0$, for $r = 1, 2, \ldots, \ell$. Of course, if the process occurs in such a fashion, we can always retrace the steps and go back from $\{P_f, V_f, T_f\}$, $\{P_i, V_i, T_i\}$. Thus, such a process is 'reversible'.

Irreversible processes are essentially every other kind of process apart from the above. Typically, one or macroscopic variables are changed abruptly in such a process. The system is thrown out-of-equilibrium, i.e, the equation of state does not hold. Then the system relaxes to a new equilibrium state again. For example, if we abruptly change $\{V_i, T_i\}$ to $\{V_f, T_f\}$, then at the instant this change is done, the macroscopic variables are $\{P_i, V_f, T_f\}$. That is to say, since the change was abruptly done, the pressure did not have time to change to a new value at that instant. Clearly, $f(P_i, V_f, T_f) \neq 0$, so equation of state does not hold, and the system is out-of-equilibrium. The system then relaxes to equilibrium, such that the pressure finally becomes P_f and the equation of state is satisfied again. There is no way to retrace this change from $\{P_i, V_f, T_f\}$ to $\{P_f, V_f, T_f\}$ backwards. This is an irreversible process.

The irreversibility of a process can be quatified in terms of the second law, i.e, Eq. (1.5). The equality in Eq. (1.5) holds only for reversible processes, while for irreversible processes, the left-hand-side is strictly greater than zero, i.e.,

$$S_f - S_i + \int_i^f \frac{-đQ}{T} = 0, \qquad \text{for reversible processes.,} \tag{1.6}$$

$$S_f - S_i + \int_i^f \frac{-đQ}{T} > 0, \qquad \text{for irreversible processes.} \tag{1.7}$$

The above statements can be explicitly proven mathematically, but that would require delving into full details of a thermodynamics course, so we will not discuss the proof. More importantly, from above, we see that for an infinitesimal reversible process, we have

$$dS = \frac{đQ}{T} \qquad \text{[for infinitesimal reversible processes, i.e, close to equilibrium]} \tag{1.8}$$

This is perhaps one of the most common definition of thermodynamic entropy in popular literature. It is important to note at that we are using dS and not $đS$. So, $\int_i^f dS = S_f - S_i$, with $S_f = S_f(P_f, V_f, T_f)$, $S_i = S_i(P_i, V_i, T_i)$. So entropy is a function of macroscopic variables of the state, much like, the internal energy E.

Partial derivatives of entropy —

We have been considering a system with macroscopic variables $\{P, V, T\}$. But, now, with some foresight into what is going to covered in these lectures, we will consider the case where number of particles can also change. So the macroscopic variables are $\{P, V, T, N\}$. With the four macroscopic variables, the equation of state is $f(P, V, T, N) = 0$. This is nothing fancy. Even the ideal gas equation $PV = nRT$ contains the number of moles n, which is nothing but N/N_A, where N_A is Avogadro's number. Now there are three independent macroscopic variables in equilibrium, since one of them can be eliminated using the equation of state.

With the possibility of N also changing, the expression for infinitesimal work done in a hydrostatic system is

$$đW = -PdV + \mu dN. \tag{1.9}$$

Here, μ is the so-called chemical potential. The first term is the work done in changing the volume of the system and the second term gives the energy required to change the number of particles. Combining Eqs.(1.9), (1.8), (1.4), we then have

$$dS = \frac{dE}{T} + \frac{P}{T}dV - \frac{\mu}{T}dN \quad \text{close to equilibrium.} \tag{1.10}$$

At equilibrium, because we have the equation of state $f(P, V, T, N) = 0$, only three of the four macroscopic variables are indepedent. We can write S and E as functions of any three of macroscopic variables, say T, V, N, i.e., $S = S(T, V, N)$, $E = E(T, V, N)$. Alternatively, we might as well express T in terms of E, by using the equation $E = E(T, V, N)$. We can use this result for T in the expression for entropy to write S as a function of E, V, N instead, i.e., $S = S(E, V, N)$. In writing this way, we choose E, V, N as the indepedent macroscopic variables, while eliminating T. Knowing that, E, V, N can be chosen as the independent macroscopic variables, i.e, any of them can be independently varied, without affecting the others, we can read Eq. (1.10) as the following sentences:

- If E is changed infinitesimally, while V and N are kept fixed, the corresponding change in entropy is proportional to $1/T$. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}. \tag{1.11}$$

- If V is changed infinitesimally, while E and N are kept fixed, the corresponding change in entropy is proportional to P/T. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}. \tag{1.12}$$

- If N is changed infinitesimally, while E and V are kept fixed, the corresponding change in entropy is proportional to $-\mu/T$. This statement is written mathematically as

$$\left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}. \tag{1.13}$$

The left-hand-side of each of Eqs. (1.11), (1.12), (1.13) is a function of E, V, N. So, Eq. (1.11) can be used to obtain a relation between E, V, N and T. Eq. (1.12) can

be used to obtain a relation between E, V, N, P, and T. From these two relations, we can eliminate E to obtain a relation among P, V, N and T, which is nothing but the equation of state. So, if we know S as a function of macroscopic variables in equilibrium, we can obtain the equation of state from its partial derivatives.

This property is not special to only entropy. In fact, any quantity that can be expressed as function of macroscopic variables in equilibrium can serve this purpose (there are not too many of them). All such quantities are called 'thermodynamic potentials'. Even the internal energy E is one such quantity.

Extensive and intensive variables —
Thermodynamics makes an important distinction between types of variables. Variables which are proportional to particle number N are called extensive variables. Variables which are independent of N are called intensive variables. Thus, we have,

$$\begin{aligned} &\text{Extensive variables: proportional to } N \rightarrow \text{ Eg} : V, E, S, N. \\ &\text{Intensive variables: independent of } N \rightarrow \text{ Eg} : P, T, \mu. \end{aligned} \tag{1.14}$$

So, if the number of particles is doubled, for example, then the extensive variables are doubled, while the intensive variables remain unchanged. Crucially, entropy is an extensive variable. This fact plays an important role in formulation of statistical mechanics.

Key takeaways:

- Macroscopic systems are well-described in terms of few macroscopic variables. For a hydrostatic system with change in number of particles being allowed, the usual macroscopic variables are pressure P, volume V, temperature T and number of particles N. A given set of values of macroscopic variables is called a macrostate of the system.

- In equilibrium, all the macroscopic variables are not independent of one another. Rather they satisfy a equation. For example, for the hydrostatic system, it can be formally written as $f(P,T,V,N) = 0$. This is equation of state. The most famous one is $PV = \frac{N}{N_A}RT$, where N_A is Avogadro's number.

- Laws of thermodynamics give the laws that the macroscopic variables follow, when they change from one set of values to another:

  First law: $đQ = dE - đW$.

  Second law:

  $$\int_i^f dS + \int_i^f \frac{-đQ}{T} = 0, \quad \text{for reversible processes.},$$
  $$\int_i^f dS + \int_i^f \frac{-đQ}{T} > 0, \quad \text{for irreversible processes.}$$

  [See below Eq. (1.4) for meaning of $đ$.]

- Reversible processes correspond to those where the system is close to equilibrium throughout the process, i.e, equation of state holds throughout the process.

- For reversible processes, i.e, close to equilibrium $dS = \frac{đQ}{T}$. So, for hydrostatic systems,

  $$dS = \frac{dE}{T} + \frac{P}{T}dV - \frac{\mu}{T}dN \quad \text{close to equilibrium.}$$

  If S is expressed as function of E, V, N, which is always possible, this gives,

  $$\left(\frac{\partial S}{\partial E}\right)_{V,N} = \frac{1}{T}, \quad \left(\frac{\partial S}{\partial V}\right)_{E,N} = \frac{P}{T}, \quad \left(\frac{\partial S}{\partial N}\right)_{E,V} = -\frac{\mu}{T}.$$

- Entropy is a thermodynamic potential. That is, knowing S at equilibrium as function of macroscopic variables, the equation of state can be obtained via taking partial derivatives. This can be, for example, done with the above equations.

- There are extensive and intensive macroscopic variables. Extensive variables are the ones that are proportional to the number of particles N, like V, E, S and N itself. Intensive variables are those that are independent of N, like P, T, μ.

1.2 The goal of statistical mechanics

As explained in previous sections, microscopic view is impractical for describing macroscopic systems. Fortunately, we can describe macroscopic systems in terms of few macroscopic variables, which obey an equation of state in equilibrium. From laws of thermodynamics, we can predict the equation of state, if we know the so-called thermodynamic potentials (like entropy) as a function of the macroscopic variables. But, we do not have a prescription to predict the thermodynamic potentials. They need to be known experimentally.

At this point, it seems the laws of macroscopic variables and microscopic variables are completely unrelated. What is the connection between them? Is there a way the thermodynamic potentials can be predicted, perhaps from a microscopic model? If so, we should be able to connect macroscopic equation of state with a microscopic description. Can we do that? The goal of statistical mechanics is to answer these questions.

Key questions to be answered by Statistical Mechanics:

- What is the relation between the microscopic view and the macroscopic view?
- Can we predict thermodynamic potentials (like entropy) from microscopic models?

2 Formulation of Statistical Mechanics

2.1 The 'law' of Statistical Mechanics

The main observations and idea —
Statistical Mechanics is formulated based on the following observations and idea.

- *Observation 2.1: There are a huge number of microstates corresponding to a given macrostate.* This is rather obvious, if you think about it. For a hydrostatic ystem of N particles, of course, various values of $\{x_j(t), y_j(t), z_j(t), p_{x_j}(t), p_{y_j}(t), p_{z_j}(t)\}$, $j = 1$ to N, must correspond to the same values of P, V, T. This is essentially a consequence of number of macroscopic variables being only few, while the number of microscopic variables being intractably large.

- *Boltzmann's equation, i.e, the 'law' of Statistical Mechanics:*

$$S = k_B \log \Omega, \tag{2.1}$$

  where Ω denotes the number of microstates corresponding to a given set of macroscopic variables. The above formula says that entropy is proportional to the logarithm of the number of microstates. It was Ludwig Boltzmann's great idea to define entropy this way, so the proportionality constant, k_B is called Boltzmann's constant.

  In every fundamental branch of Physics, we have some 'laws'. The 'laws' are not derived, but rather, just written down, may be either with experimental insight, or via some plausibility arguments. The philosophy of Physics is that, everything else can be derived from these 'laws'. We have the Newton's laws in Classical Mechanics, we have the Schroedinger equation and the postulates of measurement in Quantum Mechanics, we have the laws of Thermodynamics. Similarly, Eq. (2.1) serves as the 'law' of Statistical Mechanics. It precisely does the job of connecting a macroscopic quantity which is entropy with a quantity that explicitly depends on the microscopic description, which is the number of possible microstates.

- *Observation 2.2: S is monotonically increasing function of Ω.* If $f(x)$ is a monotonically increasing function of x it means that if x increases, $f(x)$ also increases. There cannot be a situation where x increases, but $f(x)$ decreases. So, x and $f(x)$ always increase or decrease together.

  In the language of calculus, it essentially means $\frac{df}{dx} > 0$, over the entire regime of allowed values of x. This is certainly true for the function $f(x) = \log x$, which is well-defined only for $x > 0$. The derivative would be $\frac{df}{dx} > \frac{1}{x} > 0$, for $x > 0$.

  Knowing the above Boltzmann's formula Eq. (2.1), then means that S and Ω increase or decrease together.

- *Observation 2.3: Since S is function of macroscopic variables, Ω is also a function of macroscopic variables.*

Understanding equilibrium in terms of Ω —
Consider a system completely isolated from its surroundings, for example, a gas, in a box whose walls are *perfectly insulating, so that, no heat exchange is possible with the surroundings*. Such processes are called 'adiabatic processes'. As we know, the macroscopic variables are P, V, N and T. We will consider that N and T are also kept fixed, so only P, V, can change. Now, say, initially the system is in equilibrium and the values of the macroscopic variables variables are P_i, V_i, T, N. So, the equation of state is satisfied initially, $f(P_i, V_i, T, N) = 0$.

Now, we *suddenly* change the volume of the box to V_f. At the instant of the change, the macroscopic variables are P_i, V_f, T, N, which certainly, does not satisfy the equation of state, $f(P_i, V_f, T, N) \neq 0$, so the system is out-of-equilibrium. Then the system 'equilibrates' to a new equilibrium state, with P_f, V_f, T, N, such that $f(P_f, V_f, T, N) = 0$. This process of going from the out-of-equilibrium values P_i, V_f, T, N to P_f, V_f, T, N is irreversible.

So far so good, this is very similar to the irreversible process example discussed in previous chapter. Thus, the second law of thermodynamics for irreversible processes, Eq. (1.7) holds. The only new thing we have added is that the system cannot exchange heat with the surroundings, so $đQ = 0$ throughout the process. So, from Eq. (1.7), we have

$$S_f - S_i > 0 \quad \text{for adiabatic irreversible processes.} \tag{2.2}$$

In other words, if a system, that is thermally isolated from the surroundings, 'equilibrates' starting from an out-of-equilibrium state, the entropy of the system must increase. In fact, from the second law, we know that during the entire process, since no heat is exchanged, the entropy can either remain constant for some time, or it increases, it cannot decrease. From Observation 2.2 given above, this means the number of microstates Ω also must behave similarly during the process.

So, the number of microstates, Ω, changes during the process, but cannot decrease, and at the end of the process [i.e, when the system has reached equilibrium], it becomes constant [because S becomes constant]. If you try to visualize a possible plot of how Ω can change during the process satisfying such conditions, you will realize that, when Ω becomes constant at the end, it reaches its maximum value. Therefore, we reach a very important conclusion: *the equilibrium value of* Ω *corresponds to its maximum possible value, for the given set of independent macroscopic variables.* [Note, at equilibrium, equation of state is satisfied, so number of independent macroscopic variables is one less than the total number of macroscopic variables.]

In the above example, with volume, temperature and number of particles fixed at V_f, T, N, but pressure initially being P_i, the system is out of equilibrium, $f(P_i, V_f, T, N) \neq 0$. So, Ω is not at its maximum value corresponding to the given V_f, T, N. Then the system equilibrates by changing its pressure, to P_f, such that, Ω reaches its maximum possible value for the given V_f, T, N. Thus, by maximizing Ω while keeping given values of three of the four macroscopic variables fixed, we can, in principle, determine the value of the remaining macroscopic variable, without a priori knowing the equation of state. In this way, we can obtain the relation between the four macroscopic variables in equilibrium,

or, in other words, find the equation of state.

Aside: Some curious points to note which follow from the second law, Eqs.(1.6), (1.7)*:* Without giving out heat, you cannot change a system from a state of higher entropy to a state of lower entropy. This is forbidden by the second law. Without giving out heat, you cannot perform a reversible process that changes entropy. This is also forbidden by the second law. However, without giving out heat, you can change increase the entropy of the system, via some irreversible process, like the one we discussed above. Via Observation 2.2, these statements also hold for Ω.

Key takeaways:

- A huge number of microstates, i.e, values of position and momenta of each particle, correspond to the same macrostate, i.e, some specified values of macroscopic variables.
- Boltzmann's equation is the 'law' of Statistical Mechanics:

$$S = k_B \log \Omega,$$

  where Ω is the number of possible microstates corresponding to a given set of macroscopic variables, and k_B is the, so called, Boltzmann constant.
- By Boltzmann's equation, S is a monotonic function of Ω. Also, since S is a function of macroscopic variables (say, P, V, N, T), Ω is also a function of macroscopic variables.
- Equilibrium corresponds to the maximum value of Ω corresponding to a given set of independent macroscopic variables. For example, if any three variables of a hydrostatic system (say V, T, N) are given, the equilibrium value of the remaining variable (here, P) should be such that Ω is maximized. In this way, one can find the value of the remaining macroscopic variable, without a priori knowing the equation of state.

2.2 Obtaining equation of state from a microscopic model

Using the above understanding, we can formulate a recipe to derive the equation of state starting from a microscopic model. As before, we will mainly consider hydrostatic systems, with macroscopic variables P, V, T and N. A microscopic model can have a given the number of particles N and V can be taken as the 'volume of the box' containing the hydrostatic system. So the V gives limitations on the maximum allowed values of position coordinates of the particles, $\{x_j, y_j, z_j\}$, $j = 1, 2, ..., N$. However, usually, P and T are hard to intuitively connect to a microscopic model. Rather, an expression for energy E can be written down microscopically, by just writing down an expression for

kinetic energy and potential energy,

$$E = \sum_{j=1}^{N} \left[\frac{p_{x_j}^2 + p_{y_j}^2 + p_{z_j}^2}{2m_j} \right] + \overline{V}(x_1, x_2, \ldots x_N, y_1, y_2, \ldots y_N, z_1, z_2, \ldots z_N), \tag{2.3}$$

where $\overline{V}(x_1, x_2, \ldots x_N, y_1, y_2, \ldots y_N, z_1, z_2, \ldots z_N)$ denotes the potential energy, which depends on the positions of all the particles, and m_j is the mass of the jth particle. As we have seen before, we can choose E, V and N as the three independent macroscopic variables describing our system. It therefore turns out that this choice is quite convenient to connect with a microscopic model. With this idea, the recipe to obtain the equation of state starting from a microscopic model is as given in the following box.

Recipe for deriving equation of state from microscopic model:

- Choose a volume V and number of particles N.
- Write down an expression for E in terms of sum of kinetic and potential energies. This gives the E as a function of the microscopic variables, $\{x_j, y_j, z_j, p_{x_j}, p_{y_j}, p_{z_j}\}$, $j = 1, 2, \ldots, N$.
- Count the number of possible values of $\{x_j, y_j, z_j, p_{x_j}, p_{y_j}, p_{z_j}\}$, , $j = 1, 2, \ldots, N$, that gives the same E, keeping V and N fixed. This is number of microstates Ω. Find the maximum possible value of Ω. This is the equilibrium value of Ω at the given E, V, N. This way, we obtain the equilibrium Ω as a function of the macroscopic variables, $\Omega = \Omega(E, V, N)$.
- Use Boltzmann's formula to obtain the equilibrium value of S as a function of the macroscopic variables, $S = S(E, V, N)$.
- Use the partial derivatives of $S(E, V, N)$ [Eqs. (1.11), (1.12), (1.13)] to obtain the equation of state.

If everything is correct, we should be able to write down a microscopic model, from which we can derive the ideal gas equation of state using the above procedure. This is what we will do in the next chapter.